NATURAL RESOURCES

Class 9 | Science | Chapter 14

THE BIOSPHERE

The biosphere is the zone of the Earth where life exists -- the region where land, water, and air meet and support life.

The three basic natural resources shared by all life are: Air, Water, and Soil.

THE AIR AROUND US

The atmosphere -- a blanket of air surrounding the Earth -- keeps the average temperature fairly steady and prevents sudden extreme temperature changes.

Component of Air	Approx. %
Nitrogen	78%
Oxygen	21%
Carbon dioxide, water vapour, other gases	~1%

Winds are caused by uneven heating of the atmosphere & land/water bodies by the Sun (convection currents).

AIR & WATER POLLUTION

Greenhouse Effect, Ozone & Water Cycle

AIR POLLUTION

- Caused by burning fossil fuels (coal, petroleum), industrial smoke, vehicle exhaust.
- Pollutants like CO_2, SO_2, and particulate matter (SPM) reduce air quality and harm health.

Greenhouse Effect

Gases like CO_2 trap heat radiated from Earth, warming the atmosphere -- essential in the right amount but excess causes global warming.

Ozone Layer

A layer of O_3 in the upper atmosphere that absorbs harmful UV rays from the Sun; being depleted by CFCs (chlorofluorocarbons).

WATER

Around 2/3rd of the Earth's surface is covered with water, but only a small fraction is fit for use (fresh water).

The Water Cycle

Evaporation from water bodies -> condensation into clouds -> precipitation (rain/snow) -> collection back into water bodies -- continuously recycles water.

SOIL

Formation, Types & Erosion

SOIL FORMATION

Soil is formed by the weathering of rocks -- through the action of wind, water, and living organisms -- over a very long time.

- Physical/mechanical weathering: temperature changes, wind, water flow break rocks apart.
- Chemical weathering: rainwater reacts with minerals in rock.
- Biological weathering: lichens, plant roots, and burrowing organisms break down rock further.

SOIL PROFILE / LAYERS

Layer	Feature
Topsoil (A-horizon)	Dark, rich in humus & nutrients, supports most plant roots
Subsoil (B-horizon)	Less humus, more minerals
Bedrock (C-horizon)	Weathered rock fragments, closest to parent rock

SOIL EROSION & CONSERVATION

Soil erosion = removal/wearing away of the top fertile soil layer by wind or water.

Conservation methods: afforestation (planting trees), terrace farming on slopes, building contour bunds, crop rotation.

BIOGEOCHEMICAL CYCLES & SUMMARY

Nitrogen, Carbon & Oxygen Cycles

NUTRIENT CYCLES IN NATURE

Nitrogen Cycle

Atmospheric N2 -> fixed by bacteria (Rhizobium) or lightning -> used by plants -> passed through food chain -> returned to atmosphere by denitrifying bacteria.

Carbon Cycle

CO2 taken up by plants (photosynthesis) -> passed through food chain -> returned to atmosphere via respiration, burning fossil fuels & decomposition.

Oxygen Cycle

Released by plants during photosynthesis; used by all organisms during respiration and combustion.

CHAPTER SUMMARY -- KEY TERMS

- **Biosphere** -> zone of Earth that supports life
- **Greenhouse effect** -> trapping of heat by atmospheric gases
- **Ozone layer** -> absorbs harmful UV rays
- **Water cycle** -> evaporation-condensation-precipitation loop
- **Soil erosion** -> loss of fertile topsoil

ONE-LINE RECAP FOR REVISION

Air, water and soil together support all life in the biosphere.

Natural cycles (water, carbon, nitrogen, oxygen) keep resources renewed.

Human activity (pollution, deforestation) disrupts these natural cycles.

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